School of Engineering & Technology, CGC University
Mohali
Department of Artificial Intelligence & Data Science

ENGINEERING CLINIC — PROJECT SYNOPSIS

SMART RECYCLING VENDING MACHINE

WITH AI MATERIAL RECOGNITION

AI-BASED / IoT-BASED ENGINEERING CLINIC PROJECT

AI / ML Technique	CNN-based image classification using transfer learning (MobileNetV2 backbone) for real-time material recognition
IoT Hardware Used	ESP32-CAM, IR Sensors, HC-SR04, SG90 Servo Motors, HX711 + Load Cell, 16x2 I2C LCD, RC522 RFID Reader, DC Geared Motor
Team Members	Ayush — Team Leader — 2510003099 Shubham — 123013 {GR no} Arshdeep — 2510003075 Pooja — 2510003103
Project Mentor	Mrs. Baljinder Kaur, Assistant professor Mr.HemantKushwaha , Assistantprofessor
Department	[CGCU APEX]
Academic Year	[2026-27]

TABLE OF CONTENTS

1. ABSTRACT..... 3

2. INTRODUCTION..... 3

3. PROBLEM STATEMNET..... 4

4. OBJECTIVES..... 5

5. LITERATURE SURVEY/ EXISTING SYSTEMS..... 5

6. PROPOSED SYSTEM..... 6

6.2 BLOCK DIAGRAM 7

6.3 METHOLOGY..... 7

7. HARADWARE REQUIREMENTS..... 8

8. SOFTWARE AND TECHNOLOGIES USED 9

9 EXPECTED OUTCOMES..... 10

10. APPLICATIONS 10

11. ADVANTAGES AND LIMITATIONS 11

12. FUTURE SCOPE..... 12

13. WORK PLAN / TIMELINE 12

14 REFERENCES..... 13

1 ABSTRACT

Improper segregation of recyclable waste at the point of disposal remains one of the biggest bottlenecks in modern recycling systems — mixed or contaminated waste streams are frequently rejected outright at material recovery facilities. This project proposes a Smart Recycling Vending Machine that uses AI-based material recognition to automatically identify and sort waste items such as plastic, metal, glass, and paper the moment they are deposited by a user.

An ESP32-CAM module captures an image of the discarded item, which is then classified by a Convolutional Neural Network built through transfer learning on a MobileNetV2 backbone and trained on labelled waste-image data. Based on the predicted class, the ESP32 microcontroller drives a servo-based sorting mechanism to route the item into its correct compartment, while a load cell logs its weight.

Successfully sorted items trigger a reward token for the user to encourage responsible recycling behaviour, and every activity is logged to a cloud dashboard for remote bin-level monitoring. Built around low-cost, off-the-shelf components, the system is expected to reach a high classification accuracy while staying affordableenoughfor real deployment across campuses, housing societies, and public spaces — cutting down manualsortingeffort and improvingoverall recycling rates.

2 INTRODUCTION

Solid waste generation is rising sharply with urbanisation, and a large share of recyclable material — plastic, metal, glass, and paper — never actually gets recycled, simply because it isn't segregated correctly at the source. Once recyclables get mixed with general waste, contamination makes downstream sorting expensive, energy-intensive, and often infeasible, so a lot of material that is technically recoverable still ends up in landfills. Correct segregation at the point of disposal is therefore one of the highest-leverage interventions possible in the recycling value chain.

Today, segregation depends almost entirely on the user correctly identifying which bin an item belongs to — something that stays inconsistent even with clear signage, and is genuinely hard to enforce at scale. Combining IoT sensing with AI/ML changes this: an embedded camera can observe an item the moment it is discarded, and a trained model can classify its material in real time far more consistently than relying on human judgement, while IoT connectivity lets the device report its status — bin fill-level, item counts, faults — to a remote dashboard without any manual checks.

Existing solutions really only address one half of this problem. Commercial reverse vending machines accept a narrow set of recognisable containers using barcode or shape sensing, and IoT-enabled smart bins report fill-level but do not sort anything at all. Meanwhile, published waste-classification research shows high accuracy but is mostly evaluated offline on static datasets, without ever being integrated into a working, actuated device. This project addresses that gap by combining a lightweight, camera-based CNN classifier with a real-time sorting, weighing, and reward mechanism in a single low-cost embedded system.

3 PROBLEM STATEMENT

Waste segregation is a crucial step in the recycling process, but sorting different types of recyclable waste by hand is slow and genuinely hard to sustain. In many places, plastic, metal, and paper waste all get thrown together, which makes later recycling and processing far more complicated. Manual sorting depends entirely on human effort, and mistaken segregation is common — not always, but often enough to be a real problem. There is a clear need for a simpler system that can recognise recyclable materials and move them to their own collection bins with less manual work. Many existing smart-waste-management ideas get complicated — extra hardware, multiple sensors, or heavier processing pipelines — which does not fit well within a small educational prototype. This project therefore focuses on building a low-cost automated system that can identify plastic, metal, and paper waste using AI-based image classification, and then split it automatically into

different bins, usingan ESP32-CAM for capture, a MobileNetV2-based model for recognition, and a servo mechanismfor sorting.

4 OBJECTIVES

The main objectives of the Smart Recycling Vending Machine with AI-Based Material Recognition are:

1. To design and develop a working prototype for automatic segregation of recyclable waste.
2. To use an ESP32-CAM to capture images of waste items placed inside the machine.
3. To develop an AI-based image classification system using MobileNetV2 and transfer learning to identify plastic, metal, and paper.
4. To use the classification result to control a servo motor-based sorting mechanism.
5. To provide information about the detected waste through a 16×2 I2C LCD display.
6. To include a simple reward-point system that encourages users to dispose of recyclable waste properly.
7. To evaluate the prototype on classification accuracy, sorting accuracy, response time, and overall cost.
8. To keep the prototype simple and affordable, targeting an approximate cost of ₹4,000.

5 LITERATURE SURVEY / EXISTING SYSTEMS

The table below summarises representative existing work relevant to AI-based waste classification and IoT-based sorting or collection systems.

#	Existing Work	Approach	Limitation
1	TrashNet dataset with baseline CNN/SVM classifiers [1]	~2,527 labelled images across 6 waste classes, benchmarked with CNN and SVM classifiers	Small, studio-lit dataset; accuracy drops sharply under real-world lighting and backgrounds
2	Transfer-learning waste classifiers — ResNet-50, MobileNetV2, DenseNet-121 [2]	Pretrained CNN backbones fine-tuned on waste images, reporting up to ~95% accuracy on curated test sets	Evaluated offline on static datasets; not integrated with real-time embedded hardware or physical
3	Commercial Reverse Vending Machines (RVMs)	Barcode scanning and weight/shape sensing to accept bottles and cans for deposit-refund schemes	Restricted to a narrow set of recognisable containers; cannot generalise to mixed material types
4	IoT-enabled smart bins with ultrasonic fill sensors and Wi-Fi/GSM alerts	Monitor bin fill-level and notify staff for collection scheduling	Solves the collection-logistics problem only; waste is still placed into one mixed bin Better robustness comes at the cost of heavier compute, harder to run on low-cost embedded boards
5	Federated/hybrid CNN approaches on extended datasets — TrashBox, TACO [3][4][5]	Extend TrashNet with larger, real-world image sets to improve robustness to clutter and lighting	

Table no. 1

Overall, the research gap is fairly clear: high-accuracy material classification exists mainly as an offline benchmarking exercise, while real-time sorting hardware such as RVMs and smart bins exists mainly without any material-level intelligence. This project closes that gap by embedding a lightweight CNN classifier directly into a low-cost, actuated vending-style machine that senses, classifies, sorts, weighs, and rewards within a single real-time pipeline.

6 PROPOSED SYSTEM

6.1 System Overview

The proposed system follows a sensing → recognition → decision → actuation pipeline. When a user deposits a waste item into the intake chute, an IR sensor detects its presence and triggers the ESP32-CAM to capture an image. This image is passed to a CNN-based material recognition model, transfer-learned on a MobileNetV2 backbone, which classifies the item into one of four categories — plastic, metal, glass, or paper. The predicted class is sent to the central ESP32 microcontroller, which drives a servo-controlled sorting flap and a DC geared motor (turntable/conveyor) to route the item into the corresponding compartment.

A load cell under each compartment logs the added weight, and an HC-SR04 ultrasonic sensor tracks each bin's fill-level. Once the item is confirmed sorted, the user taps an RC522 RFID card to identify their account, the system displays a confirmation on the LCD, and a reward point/token is credited. Bin weight and fill-level data are periodically pushed to a cloud dashboard so facility staff can monitor status and schedule collection remotely.

6.2 Block Diagram

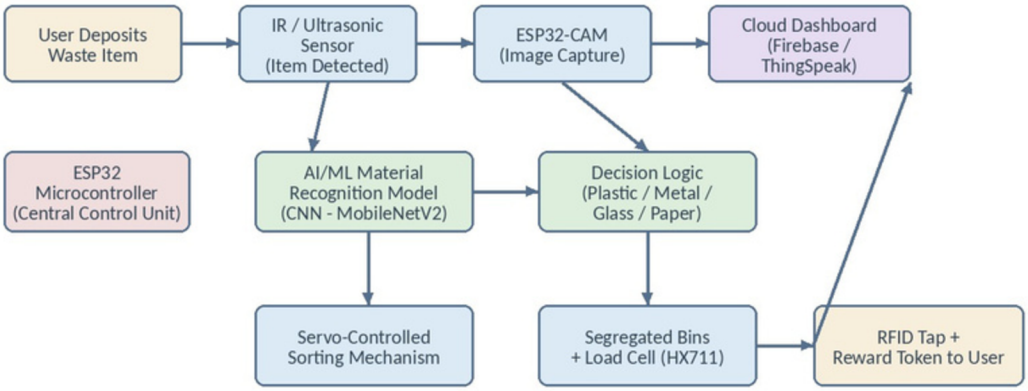


Figure 1: System Block Diagram of Smart Recycling Vending Machine with AI Material Recognition

6 PROPOSED SYSTEM — METHODOLOGY / WORKING PRINCIPLE

6.3 Methodology/Working Principle

1. Data acquisition: the IR/ultrasonic sensor detects item presence at the intake; the ESP32-CAM captures a single top-down image once per deposit event.
2. Preprocessing: the captured image is resized to 224×224 pixels, normalised, and background-cropped before being passed to the model.
3. AI/ML model: a MobileNetV2 backbone pretrained on ImageNet is fine-tuned via transfer learning on a labelled waste-image dataset (TrashNet, augmented with locally collected images), chosen for its balance of accuracy and low inference latency suited to near-edge deployment; the output is a 4-class probability vector.
4. Decision logic: the class with the highest softmax confidence is accepted if it crosses a set threshold (e.g., ≥70%); predictions below the threshold are flagged “unidentified” and routed to a general reject bin instead of being mis-sorted.
5. Output/action: the ESP32 actuates the corresponding servo flap and DC geared motor, the load cell logs weight, the user taps their RFID card to identify their account, the LCD confirms the result and reward, and the event is logged to the cloud dashboard.

7 HARDWARE REQUIREMENTS

Component	Specification	Purpose in this Project	Approx. Price
ESP32	Dual-core, Wi-Fi + BLE, OV2640 2MP camera	Captures the item image, runs the classification trigger, and acts as the central controller	₹293–₹381
IR Sensor ×2	Infrared proximity sensor	Detects item presence at the intake and item passage into the sorted bin	₹60
HC-SR04	Range 2–400 cm	Measures bin fill-level for cloud reporting	₹99
HX711	24-bit ADC	Interfaces the load cell readings to the ESP32	₹41
Load Cell	5 kg capacity	Measures the weight of the sorted item per compartment	~₹100–150
SG90 Servo ×2	180° rotation, 4.8–6 V	Actuates sorting flaps to route the item into the correct bin	~₹180–200
16×2 I2C LCD	I2C interface	Displays status, classification result, and reward confirmation	₹133
RC522 RFID + Card	13.56 MHz	Identifies the user's account for logging reward points	₹58
DC Geared Motor	Low-RPM geared motor	Drives the sorting turntable/conveyor that moves the item into its bin	₹44+
Jumper Wires	M-M / M-F / F-F	Circuit connections across sensors, motors, and controller	₹73
5V 2A Supply	Regulated	Powers the ESP32, servos, motor, sensors, and display	₹80
Breadboard	Standard prototyping board		₹80–100
Compartmented Bin Enclosure	4 compartments (fabricated)		—

The itemised components above total approximately ₹1,300–₹1,500. Including wiring, switches, LEDs, and the fabricated bin enclosure/mechanism, the estimated total prototype cost works out to roughly ₹1,500–₹2,000.

8 SOFTWARE & AI/ML STACK

- Programming languages: Python for model training and evaluation; C/C++ for ESP32 firmware via Arduino IDE / PlatformIO.
- ML framework: TensorFlow / Keras for building the model, with TensorFlow Lite used for near-edge inference.
- Model architecture: MobileNetV2 with transfer learning, fine-tuned for 4-class material classification.
- Dataset: TrashNet used as the baseline [1], augmented with locally captured images specific to the project's four target classes.
- Supporting tools: Arduino IDE / PlatformIO for firmware; the MFRC522 library for RFID reader integration; EdgeImpulse (optional)foron-deviceoptimisation; Firebase Realtime Database or ThingSpeak for cloud

9 EXPECTED OUT COME S

logging;and a lightweightwebdashboard(HTML/JS with Chart.js) for visualisation.

After completing the project, the goal is a working prototype of a Smart Recycling Vending Machine capable of identifying and sorting three common recyclable materials — plastic, metal, and paper. In particular, the project is expected to deliver:

1. A functional prototype for automatic waste segregation.
2. An AI model capable of classifying plastic, metal, and paper waste.
3. Automatic sorting of the detected material using a servo motor.
4. Reliable detection of incoming waste using an IR sensor.
5. Clear display of the detected material and reward points on the LCD.
6. A target classification accuracy of around 85% or above on test data, depending on factors such as lighting, camera quality, and object placement.
7. A prototype cost targeted at around ₹4,000.

10 APPLICATIONS

The proposed system can be useful in a number of settings where basic recyclable-waste segregation is needed.

1. Educational Institutes — the machine can be used in colleges and schools to demonstrate the practical use of AI, IoT, image processing, and embedded systems.
2. Recycling Stations — it can serve as a basic automated system for separating recyclable materials before further recycling.
3. Smart Campus — the system can be installed across college campuses to encourage students to dispose of recyclable waste properly.
4. Shopping Malls and Public Places — a more advanced version could be deployed in public areas where large numbers of people generate recyclable waste.
5. Smart City Projects — the underlying concept can be extended and integrated into larger waste-management systems used across smart cities.

11 ADVANTAGES & LIMITATIONS

Advantages

- Removes dependence on the user's own knowledge of, or willingness to follow, correct sorting practice.
- Provides an incentive — the reward token — that can genuinely improve recycling participation.
- Real-time cloud monitoring reduces unnecessary bin-collection trips for facility staff.
- Modular design, so additional material classes or sensors can be added later without redesigning the system.

Limitations

- Classification accuracy depends on lighting conditions and item orientation; occluded or wet items may be misclassified.
- The current prototype handles one item at a time across a fixed set of four material categories.
- Periodic recalibration or retraining will be needed as new packaging materials appear.
- Cloud logging depends on Wi-Fi connectivity; offline operation comes with reduced monitoring functionality.

12 FUTURE SCOPE

The project leaves plenty of room for improvement beyond this initial prototype. In future iterations, the AI model can be trained to recognise more waste categories, such as glass, cardboard, e-waste, and other recyclables, and the sorting mechanism can be made sturdier and more reliable for objects of different sizes and shapes.

A companion mobile application could also be built to maintain user accounts and track reward points earned through recycling, making the reward system more meaningful and encouraging people to recycle on a regular basis.

Other additions, such as bin fill-level monitoring with alerts when a compartment is nearly full, would also be valuable.Withcontinued improvementsin hardware, AI, and monitoring, this prototype could eventually grow intoagenuinelypractical systemforlarger recycling facilities or smart-city deployments.

13 WORKPLAN / TIMELINE

Phase	Weeks	Deliverable
Literature review, requirement analysis & component procurement	Weeks 1–2	Finalised component list, literature survey report
Hardware assembly (sensor, camera, servo, load cell integration) & dataset collection	Weeks 3–4	Working sensor/actuator rig, local image dataset
Model training & evaluation (transfer learning, threshold tuning)	Weeks 5–6	Trained CNN model with accuracy report
Firmware-model integration & cloud dashboard setup	Weeks 7–8	End-to-end functional prototype
System testing, refinement & documentation	Week 9	Test report, bug fixes
Final demo, report & presentation preparation	Week 10	Final synopsis/report, working demo

Table no.2

14 REFERENCES

[1] G. Thung and M. Yang, "trashnet: Dataset of images of trash for garbage classification," GitHub repository, 2016. [Online]. Available: https://github.com/garythung/trashnet

[2] "Classification of Waste Materials using CNN Based on Transfer Learning," ACM Digital Library, 2023. [Online]. Available: https://dl.acm.org/doi/fullHtml/10.1145/3574318.3574345

[3] "Enhancing trash classification in smart cities using federated deep learning," Scientific Reports, 2024. [Online]. Available: https://www.nature.com/articles/s41598-024-62003-4

[4] "Towards artificially intelligent recycling: Improving image processing for waste classification," arXiv:2108.06274, 2021.

[5] "Towards sustainable solutions: Effective waste classification framework via enhanced deep convolutional neural networks," PLOS ONE, 2025. [Online]. Available: https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0324294

[6] Espressif Systems, "ESP32-CAM Series Datasheet," 2023.

[7] SparkFun Electronics, "HX711 Load Cell Amplifier Datasheet."